

The first exam is on **Friday, September 24, 2021**. Please bring pencil (0.7 mm-0.9mm), non-programmable calculator (**NCEES**), and eraser with you. Exam is closed book and closed notes. You must show the full details of your work for full credit. **No credit just for answer!** Storing the formulas or notes related to EMCH 290 is absolutely prohibited in the calculator. Please turn off your cell phone and laptop and keep them in your backpack. It is the responsibility of every student at the University of South Carolina-Columbia to adhere steadfastly to truthfulness and to avoid dishonesty, fraud, or deceit of any type in connection with any academic program. Any student who violates this Honor Code or who knowingly assists another to violate this Honor Code shall be subject to discipline.

You have **50 minutes** to finish your Exam (**9:40 AM – 10:30 AM**). Make-up exam will not be given unless by the approval of Mechanical Engineering Department. Please make sure to turn in your exam to me at the end of exam period regardless how you did! Missing exam will be considered **zero**. **If you arrive late, you will NOT be given extra time to complete the test.** **The students with Disability should reserve a seat a week before the date of exam at Disability Office with extended time.**

The exam # 1 will cover First Law Chapter as it was discussed previously. Please make sure that you will understand the following materials:

- a. Dimensions and Units (SI Units, English Engineering System)
- b. Work, Kinetic Energy, and Potential Energy**
- c. Units of Energy in SI System and English Engineering System.
- d. Power, sign convention and notation for work and heat.
- e. Expansion or compression work in process (example 2.1 p. 50 of reference book).
- f. Energy Transfer by Heat
- g. Energy Concepts, Energy Balance Equation for closed system (First Law of Thermodynamics).
- h. Processes of closed systems such as cooling a gas in a piston-cylinder (example 2.2 & 2.3).
- i. Energy Analysis of Cycles, Cycle Energy Balance (Power Cycles, Refrigeration Cycle).
- j. Thermal Efficiency ( $\eta$ ), Coefficient of Performance for refrigeration ( $\beta$ ), and Coefficient of Performanc.

\*The Exam is reasonable and easy if you understand the covered materials in the class, homework, and my notes.

\*\*All Conversion Tables and Required Equations will be provided during exam.